Chemistry Practice Questions

Compiled Study Materials

I. Mole Concept & Stoichiometry

1. Mass of 2×10^{21} number of atoms of an element is 0.4g. What is the mass of 0.5 mole of the element?

2. How many numbers of moles of CO will be left when 2×10^{21} molecules are removed from 0.28g CO?

3. Calculate mass in gram:

   a) Two atoms of Carbon

   b) Three molecules of Hydrogen

4. How many grams of sulphur and oxygen are needed to produce 6.022×10^{24} molecules of SO_2 as per reaction:

$$S + O_2 \rightarrow SO_2$$

II. Limiting Reagents & Chemical Reactions

1. 2g of Magnesium is burnt in a closed vessel containing 1.2g of oxygen to produce magnesium oxide.

   a) Find limiting reagent.

   b) Calculate unreacted no. of molecules of reagent left over.

   c) What mass of MgO is produced?

   d) How many grams of pure HCl are needed to neutralize whole MgO produced?

2. Consider a reaction: $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$. 10g of Fe_2O_3 is reacted with 9g of CO.

   a) Find the limiting reagent.

   b) How many moles of unreacted reactant left over?

   c) Calculate no. of mole of CO consumed.

   d) What mass of pure NaOH is required to absorb whole CO_2 produced in the reaction?

III. Gas Laws, Purity & Dilution

1. For a reaction: $2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$. 2.5g of a sample $NaHCO_3$ when strongly heated gives 310 cc of CO_2 gas at 27°C and 760 mm of Hg pressure.

   a) Calculate % purity of $NaHCO_3$.

   b) How many moles of H_2O are produced?

   c) What mass of pure HCl is required to neutralize Na_2CO_3 produced in the reaction?

2. How much sulphuric acid containing 80% of H_2SO_4 by weight is needed to produce 500 kg of 80% HCl by weight?

$$2NaCl + H_2SO_4 \rightarrow 2HCl + Na_2SO_4$$

3. 73g of conc HCl was diluted by adding 144g of water. How many gram atom of Hydrogen are present in dilute acid?

IV. Atomic Structure & Periodic Trends

1. Ar comes before K in the periodic table, but Ar has a larger relative atomic mass than K. Explain.

2. Is a potassium atom larger, smaller or the same size as a potassium ion? Explain.

3. Which of the following pairs would have a larger size and why?

   (a) K or K^+

   (b) F or F^-

4. Why do metals form positive ions and non-metals form negative ions?

5. Why is the size of Cl^- ion is larger than Cl atom where as size of K^+ ion is smaller than that of K atom?

6. Define ionization energy. How magnitude of ionization energy? Ionization energy of 'N' is higher than that of 'O'. Give reason.

7. Predict which of the following pair has larger electron affinity and why? O and F.

8. Why does the first Ionization energy increase from left to right in a given period of the periodic table?

9. Why ionization energies of alkalis metals decreases as the atomic number increases.

10. Arrange the elements Na, Li and K in the Increasing order of first ionization energy.

11. Which one N or O has more electron affinity and why?

V. Oxides & Allotropes

1. Choose oxide from the following that you would expect to be acidic. Give reaction to justify. Your choice Na_2O, ZnO, MgO, SO_2.

2. Name the allotropes of oxygen. Give an important use of each.

3. What are oxides? Classify the following oxides justifying the classification:

   (a) NO (b) CO_2 (c) ZnO (d) CaO

4. Classify the following oxides: N_2O_3, NO, BaO_2 and Fe_2O_3.

5. Give reason:

   a. CO_2 as an acidic oxide

   b. Al_2O_3 as an amphoteric oxide

6. Classify the following oxides: N_2O_3, NO, BaO_2, Fe_3O_4.

7. Classify the following oxides with reason: Al_2O_3, CO_2.

VI. Organic Chemistry: Isomerism & Effects

1. Define Isomerism.

2. Short Note on Structural Isomerism.

3. Define functional isomerism.

4. Write down isomers of C_6H_{14} with their IUPAC Name.

5. Give functional isomers of ethanoic acid and propanone with structural formula.

6. Write down functional isomer of:

   a. $CH_3CH_2CH_2OH$

   b. CH_3CH_2COOH

   c. C_3H_6O

   d. C_3H_7OH

7. Define Inductive Effect giving an example of it and write its one use.

8. What are electrophiles? Write any two examples.

9. Define Nucleophiles giving example.

10. How is an electrophile differed from nucleophile. Give a suitable example of each?

VII. Organic Conversions & Reactions

1. Convert sodium benzoate to Benzene Hexachloride.

2. What happens when
   i. Sodium Benzoate is heated with sodalime.
   ii. Chlorobenzene is heated with $LiAlH_4$.

3. Convert Ethyne into BHC.

4. Identify A and B:
Chlorobenzene → ($LiAlH_4/\Delta$) → A → ($Ni/H_2/\Delta$) → B

5. Identify A and B:
Sodium benzoate → ($NaOH+CaO/\Delta$) → A → (H_2/Ni) → B

6. Identify A and B:
A → ($NaOH/CaO/\Delta$) → B → (Cl_2/light) → $C_6H_6Cl_6$

7. Identify A and B:
CaC_2 → (H_2O) → A → (Red hot Cu/Δ) → B

8. Identify A and B:
Chlorobenzene → ($LiAlH_4/\Delta$) → A → (CH_3Cl/Δ / $AlCl_3$) → B

9. State Huckel's rule for Aromaticity. Why Benzene is aromatic?

10. Write resonance hybrid structure of arene containing meta director and ortho director substituent of each.

11. Example of Friedel Craft's Alkylation.

12. Benzene is heated with acetic anhydride in presence of anhydrous $AlCl_3$.

13. Show your acquaintance with Friedel Craft Reaction.